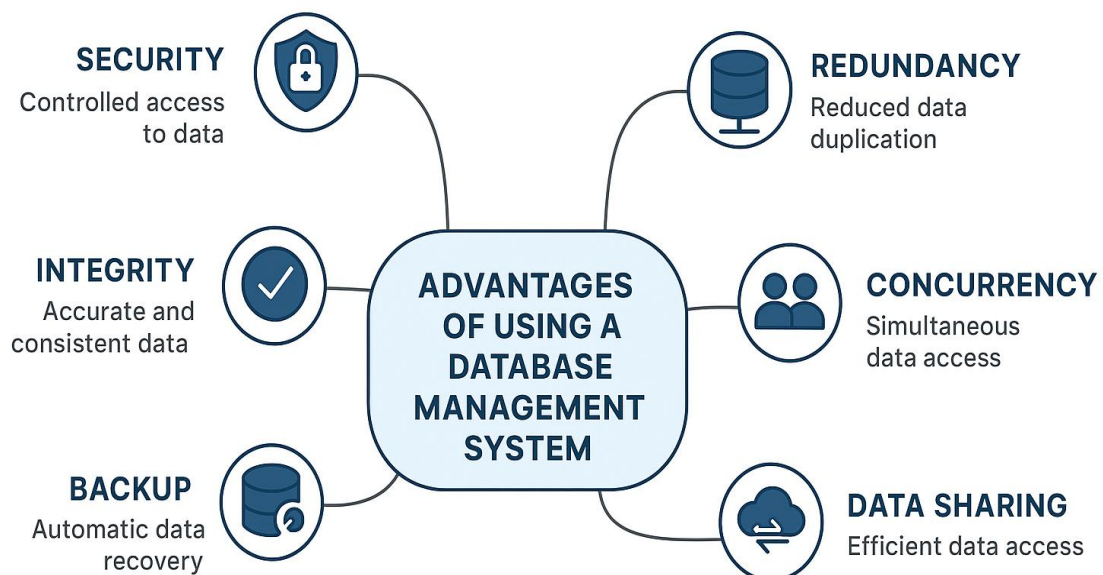


1. Comparison Assignment: Flat File Systems vs. Relational Databases

Aspect	Flat File Systems	Relational Databases
Structure	Data is stored in a single table or plain text file (e.g., CSV). Each line represents one record, with fields separated by commas or tabs.	Data is stored in multiple tables with rows and columns. Each table has a primary key , and tables are linked using foreign keys .
Data Redundancy	High redundancy — duplicate data often appears across files.	Low redundancy — data is normalized and stored once, reducing duplication.
Relationships	No built-in relationships: any links between data must be handled manually.	Tables are related to using keys, allowing easy joints and queries across multiple tables.
Example Usage	- Simple data storage (CSV, text files) - Small apps or backups. - Configuration or log files.	- Enterprise systems (HR, hospital, banking. - Websites and business applications - Reporting and analytics systems
Drawbacks	- Hard to update or search large datasets. - No validation or relationships. - Risk of data inconsistency.	- More complex to set up and maintain - Requires a DBMS (e.g., MySQL, Oracle) - Can be slower for small tasks

2. DBMS Advantages – Mind Map



3. Roles in a Database System

Role	Main Role / Focus	Typical Responsibilities
System Analyst	Bridge between business users and technical team	- Gather and analyze user requirements - Define system objectives and specifications - Create workflow diagrams - Ensure the database meets business needs
Database Designer	Plan the structure and layout of the database	- Create data models and ER diagrams - Define tables, relationships, and constraints - Ensure data integrity and efficiency - Collaborate with analysts and developers
Database Developer	Build and implement the database structure	- Create tables, views, and stored procedures - Write and optimize SQL queries - Implement relationships and indexing - Test and debug database features
Database Administrator (DBA)	Manage, maintain, and secure the database system	- Install and configure DBMS software - Monitor performance and security - Handle backups and recovery - Ensure high availability and reliability
Application Developer	Develop applications that interact with the database	- Connect software to the database - Build user interfaces and reports - Validate and update data - Optimize app–database interaction
BI (Business Intelligence) Developer	Analyze data and create business insights	- Design dashboards and reports - Build ETL processes and data warehouses - Use BI tools (e.g., Power BI, Tableau) - Support data-driven decision-making

4. Types of Databases

- Relational vs. Non-Relational Databases

Type	Description	Example	Use Case
Relational Database	Stores data in structured tables with rows and columns. Uses SQL and supports relationships between tables through primary and foreign keys	- MySQL - Oracle - PostgreSQL - Microsoft SQL Server	Best for structured data such as employee records, financial data, or inventory systems.
Non-Relational Database	Also called NoSQL; stores unstructured or semi-structured data (documents, key-value pairs, graphs). No fixed schema.	- MongoDB - Cassandra - Redis	Ideal for handling big data, social media content, IoT data, or real-time analytics.

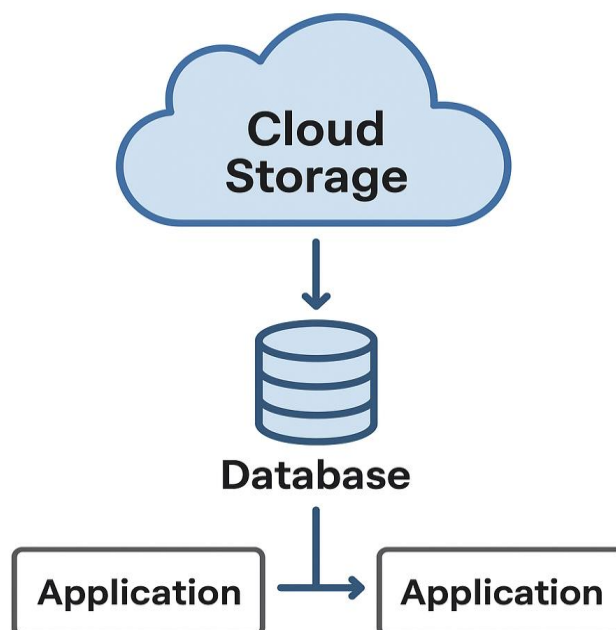
- Centralized vs. Distributed vs. Cloud Databases

Type	Description	Example	Use Case
Centralized Database	All data is stored and managed in a single location or server . Users access it remotely.	- Microsoft Access - IBM Db2	Small organizations with limited users; simple data management systems.
Distributed Database	Data is spread across multiple locations or servers but acts as a single system .	- Google Spanner - Apache Cassandra	Large companies with global operations need faster local access and reliability.
Cloud Database	Hosted on cloud platforms, accessed via the internet. Offers scalability, flexibility, and pay-as-you-go pricing.	- Amazon RDS - Google Cloud SQL - Microsoft Azure SQL	Businesses needing remote access, scalability, and minimal hardware management.

5. Cloud Storage and Databases

1. What is Cloud Storage and How It Supports Database Functionality?

- **Cloud Storage** is an online service that allows users and organizations to **store, manage, and access data** over the internet instead of on local servers or computers.
- It provides **scalable, on-demand storage** offered by cloud providers such as Amazon Web Services (AWS), Microsoft Azure, and Google Cloud.
- Support for Databases:
 - Cloud storage serves as the **foundation** for cloud-based databases.
 - Databases use cloud storage to save tables, logs, backups, and large datasets.
 - It allows databases to scale easily, replicate data across regions, and remain accessible from anywhere.
- Example:
 - When using **Amazon RDS**, the database is hosted on AWS cloud servers, and the underlying storage (Amazon S3 or EBS) securely stores the actual data.



2. Advantages of Using Cloud-Based Databases

- **Scalability:** Databases can automatically scale up or down based on demand, without manual hardware upgrades.
 - **Accessibility:** Data can be accessed securely from anywhere with an internet connection.
 - **Cost Efficiency:** Pay-as-you-go pricing models no need to buy physical servers or maintenance equipment.
 - **Automatic Backups:** Cloud providers offer built-in backup and recovery options, reducing data loss risk.
 - **High Availability:** Data is replicated across multiple regions, ensuring up time and disaster recovery.
 - **Maintenance-Free:** The cloud provider handles software updates, patches, and performance monitoring.
- **Examples:**
- **Azure SQL Database** – fully managed relational database on Microsoft Azure.
 - **Google Cloud Spanner** – globally distributed and highly scalable relational database.

3. Disadvantages or challenges with cloud-based databases

- **Internet Dependency:** Requires a stable internet connection to access or manage data.
- **Data Security & Privacy Concerns:** Sensitive data may face security risks if not properly encrypted or configured.
- **Vendor Lock-In:** Moving databases between cloud providers can be difficult and costly.
- **Cost Over Time:** Long-term cloud usage can become expensive, especially with large-scale data storage.
- **Limited Control:** Users rely on the cloud provider for updates, storage management, and configurations.